

Spatiotemporal Modeling of Flood-Associated Leptospirosis Using an Environmental SEIR Model

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Abstract

Leptospirosis is a predominantly environmentally transmitted disease, frequently associated with flooding and contact with contaminated water or mud. This study presents a mathematical-computational model to simulate its temporal and spatial dynamics after flood events. The approach integrates an environmental SEIR model with a two-dimensional cellular automaton. Baseline, severe-flood, personal-protection, environmental-decontamination, and combined-intervention scenarios were compared. The simulations assess cumulative symptomatic cases, peak symptomatic prevalence, time to peak, contaminated area, and box-counting dimension. The model is exploratory and educational, enabling investigation of how exposure intensity and different interventions modify the magnitude and spatial distribution of simulated outbreaks.

Keywords: leptospirosis; epidemiological modeling; SEIR; cellular automata; floods; environmental contamination.

1. Introduction

Leptospirosis is an infectious disease caused by bacteria of the genus *Leptospira* and is mainly transmitted through contact with water, mud, or soil contaminated by the urine of infected animals. Heavy rainfall and flooding increase this risk by spreading contaminated materials across urban areas, particularly where drainage, sanitation, and solid-waste management are inadequate [1,3].

Disease occurrence is temporally and spatially heterogeneous because it depends on rainfall intensity, persistence of environmental contamination, and local environmental characteristics. SEIR-type compartmental models represent the temporal evolution of susceptible, exposed, symptomatic, and recovered populations, while an environmental reservoir incorporates indirect exposure to the bacterium. However, such models generally do not represent local differences. Cellular automata can address this limitation by arranging individuals and contamination on a spatial grid governed by local transition rules [4,6].

This study develops an integrated environmental SEIR and cellular-automaton model to examine how flood intensity and protective or decontamination measures alter the magnitude and spatial distribution of a simulated leptospirosis outbreak. Five scenarios were compared using epidemiological indicators and the box-counting dimension of contaminated areas.

2. Methods

2.1 General framework

An exploratory Python model was developed using two complementary representations: a deterministic compartmental system for temporal dynamics and a cellular automaton for spatial distribution. NumPy, SciPy, Pandas, and Matplotlib were used. Parameters were not calibrated to a specific location; therefore, the results represent simulated scenarios rather than epidemiological forecasts.

2.2 Environmental SEIR compartmental model

The population was divided into susceptible S , exposed E , symptomatic I , and recovered R individuals, with C representing environmental contamination. The force of exposure was defined as:

$$\lambda(t) = \beta \frac{C(t)}{K + C(t)}$$

The temporal model was described by the following system of ordinary differential equations:

$$\begin{cases} \frac{dS(t)}{dt} = -\lambda(t)S(t), \\ \frac{dE(t)}{dt} = \lambda(t)S(t) - \sigma E(t), \\ \frac{dI(t)}{dt} = \sigma E(t) - \gamma I(t), \\ \frac{dR(t)}{dt} = \gamma I(t), \\ \frac{dC(t)}{dt} = \alpha + \phi F(t) - \eta C(t). \end{cases}$$

Parameters β , σ , γ , α , ϕ , and η represent environmental exposure, progression to symptoms, recovery, baseline contamination, flood effect, and decontamination, respectively. $F(t)$ is a pulse active during the flood period. Person-to-person transmission was not considered because environmental contact is the dominant transmission pathway represented in the model.

2.3 Cellular automaton representation

Space was represented by an $L \times L$ grid. Each cell contains an epidemiological state (empty, susceptible, exposed, symptomatic, or recovered) and a contamination level between 0 and 1. Environmental contamination changes according to degradation, diffusion among neighboring cells, local environmental risk, and flooding:

$$C_{i,j}(t+1) = \min \left\{ 1, \max \left[0, (1 - \eta)C_{i,j}(t) + D \left(\bar{C}_{N_{i,j}}(t) - C_{i,j}(t) \right) + \alpha r_{i,j} + \phi F_{i,j}(t) \right] \right\}$$

The transition probabilities for susceptible, exposed, and symptomatic individuals were:

$$P(S \rightarrow E) = 1 - \exp [-\beta v_{i,j} C_{i,j}(t) \Delta t],$$

$$P(E \rightarrow I) = 1 - \exp(-\sigma \Delta t),$$

$$P(I \rightarrow R) = 1 - \exp(-\gamma \Delta t).$$

Transitions were evaluated probabilistically for each individual. A Moore neighborhood was used, and all cells were updated synchronously. A fixed pseudorandom seed ensured reproducibility across repeated executions.

2.4 Simulation scenario

Scenario	Model modification
Baseline	Baseline parameters
Severe flood	+50% in ϕ and longer flood duration
Personal protection	40% reduction in β
Decontamination	+50% in η
Combined intervention	Reduction in β and increase in η

Scenarios were compared using cumulative symptomatic cases, peak symptomatic prevalence, day of peak, maximum contaminated area, and percentage change relative to the baseline scenario.

2.5 Spatial dimension

The spatial distribution of contamination was evaluated using the box-counting dimension. The grid was binarized by classifying cells above a contamination threshold C_{limit} as contaminated. For different box sizes ε , the number $N(\varepsilon)$ of occupied boxes was counted, and the dimension was estimated as:

$$D_B \approx \frac{\Delta \log N(\varepsilon)}{\Delta \log(1/\varepsilon)}$$

3. Results

In the baseline scenario, the compartmental model estimated 6,607 cumulative symptomatic cases, with a peak of 1,634 individuals on day 36. Severe flooding increased cases by 15.7%, whereas personal protection and decontamination reduced them by 27.7% and 18.0%, respectively. The combined intervention had the largest effect, reducing cumulative cases to 3,742 and peak prevalence to 978 individuals, corresponding to a 43.4% decrease relative to baseline.

The cellular automaton showed a similar pattern. Under baseline conditions, 2,378 cumulative cases and a peak of 885 symptomatic individuals on day 24 were recorded. The combined intervention reduced these values to 1,842 cases and a peak of 643 individuals, a 22.5% reduction. Severe flooding produced the largest contaminated area, reaching 3,574 cells, whereas decontamination and the combined intervention limited it to 3,234 cells. At the epidemiological peak of the baseline scenario, the box-counting dimension was estimated at 1.724, indicating an extensive and heterogeneous spatial distribution of contamination.

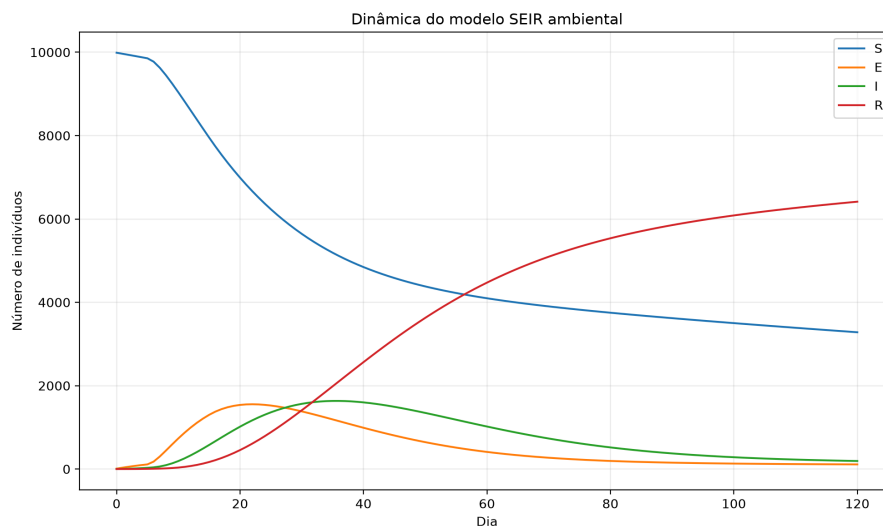


Figure 1. Temporal dynamics of the environmental SEIR model in the baseline scenario.

The curves represent susceptible (S), exposed (E), symptomatic (I), and recovered (R) populations over 120 days. Symptomatic prevalence peaked on day 36, followed by a progressive decline in active cases and an increase in recovered individuals.

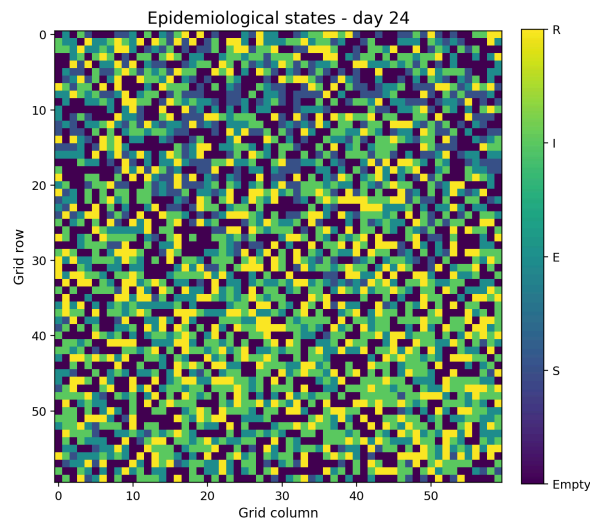


Figure 2. Spatial distribution of epidemiological states in the cellular automaton on day 24.

The map corresponds to the peak of symptomatic individuals in the baseline scenario. Cells represent empty positions or susceptible (S), exposed (E), symptomatic (I), and recovered (R) individuals, illustrating the heterogeneous spatial distribution of the outbreak.

3.1 Comparison across scenarios

Population N=10,000 in the compartmental model and N=2,508 in the cellular automaton.

Scenario	Cumulative cases - SEIR	Peak - SEIR	Cumulative cases - CA	Peak - CA	Case effect - SEIR	Case effect - CA
Baseline	6,607	1,634	2,378	885	—	—
Severe flood	7,641	1,896	2,493	991	↑ 15.7%	↑ 4.8%
Personal protection	4,775	1,116	2,132	728	↓ 27.7%	↓ 10.3%
Decontamination	5,417	1,469	2,198	820	↓ 18.0%	↓ 7.6%
Combined intervention	3,742	978	1,842	643	↓ 43.4%	↓ 22.5%

Table 1. SEIR: deterministic compartmental model; CA: cellular automaton; ↑ increase relative to baseline; ↓ reduction relative to baseline.

4. Code availability and use of artificial intelligence

The code, parameters, results, and artificial-intelligence prompts used in this study are available in the [GitHub repository](#). Artificial intelligence supported the textual structuring of the article and the initial organization of the computational implementation.

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